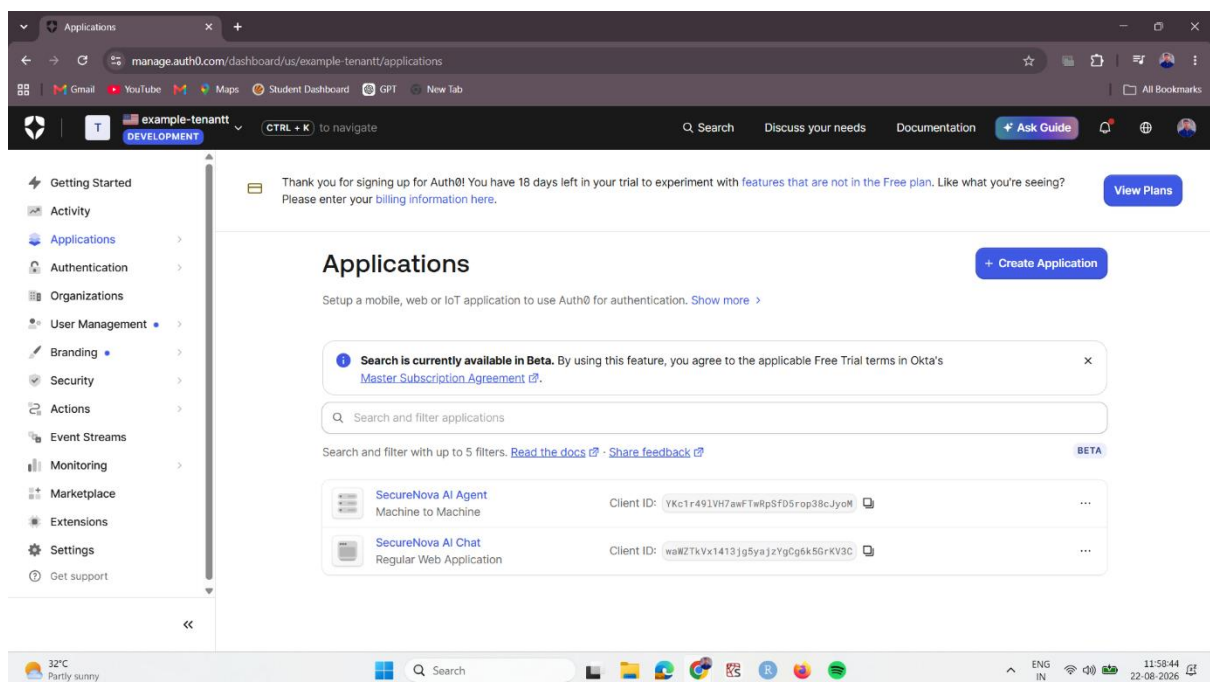


Project 2 — Build and Secure the Auth0-Backed AI Application

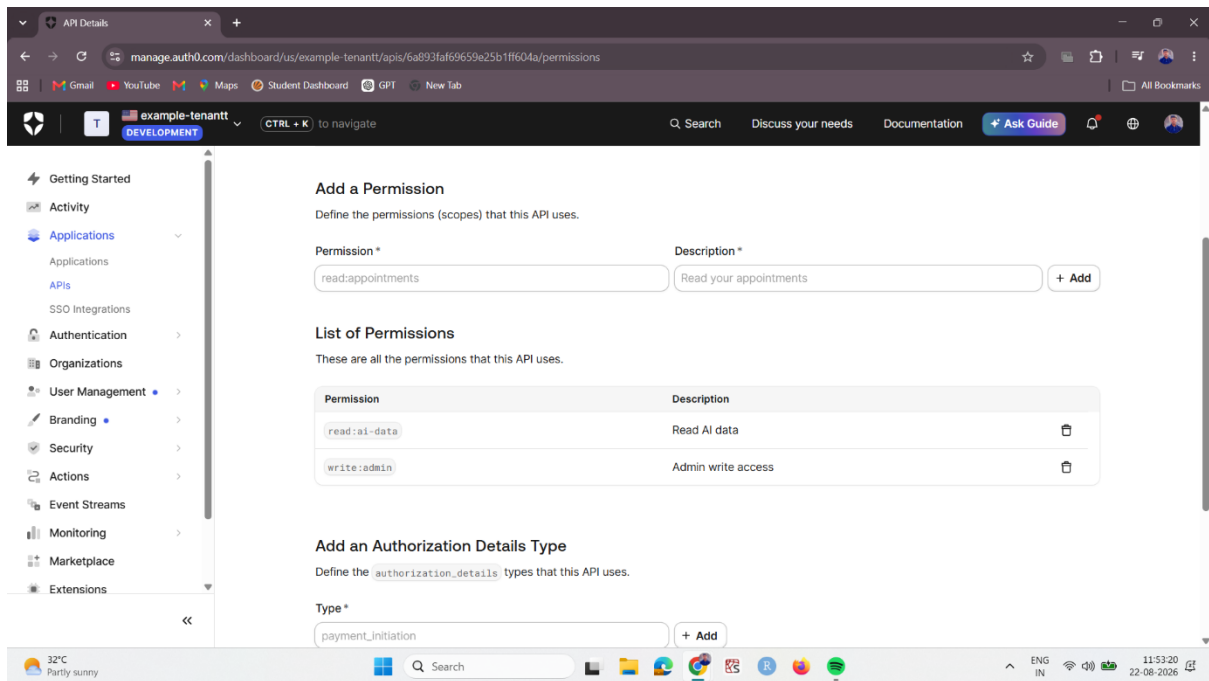
Screenshot 1 — Auth0 Applications: Web and M2M Applications

Auth0 Applications dashboard showing both the Regular Web Application for the AI chat service and the Machine-to-Machine application for the AI agent successfully registered.



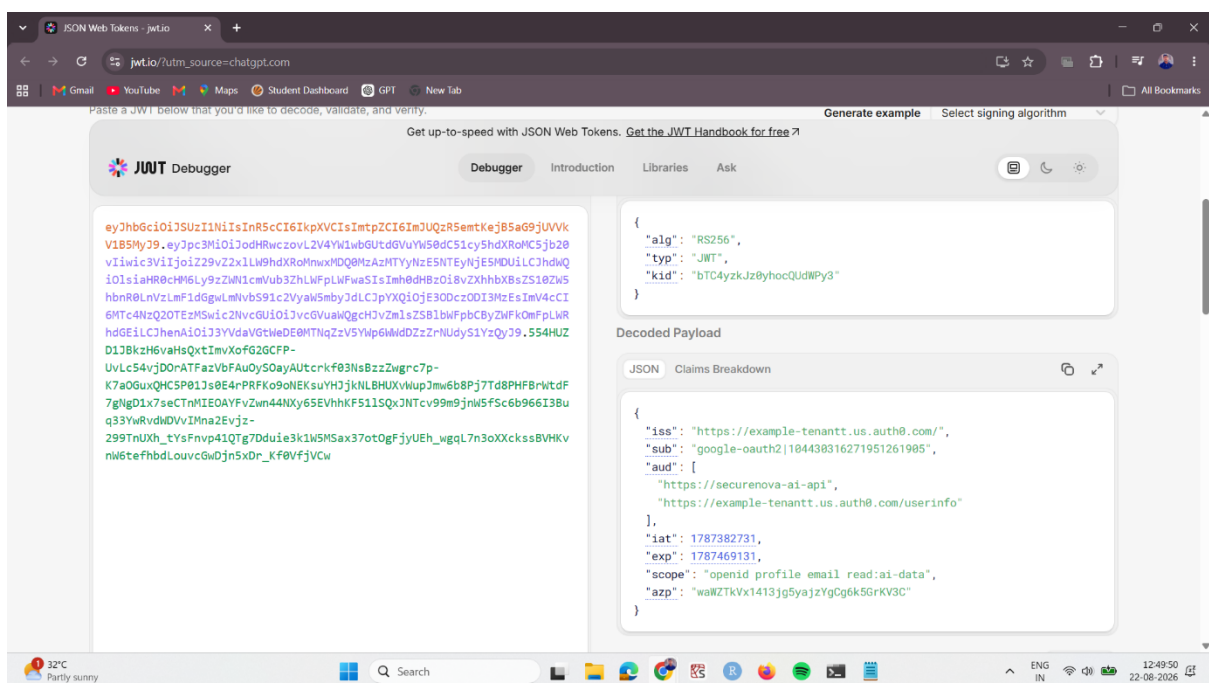
Screenshot 2 — Auth0 API: Custom Scopes

Auth0 API configuration showing the registered API with the two required custom scopes: read:ai-data for user-level access and write:admin for admin-level access.



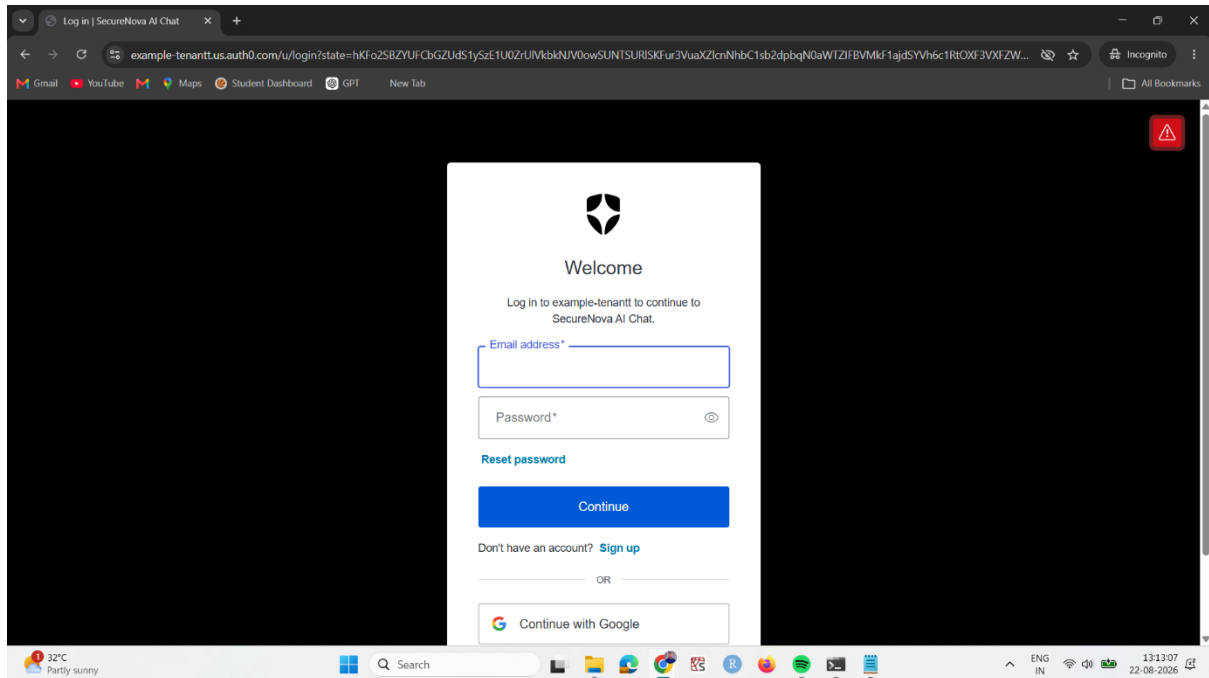
Screenshot 3 — jwt.io: Decoded JWT Claims

Decoded JWT displayed in jwt.io with the required aud, iss, exp, scope, and agent_id claims visible for token verification.



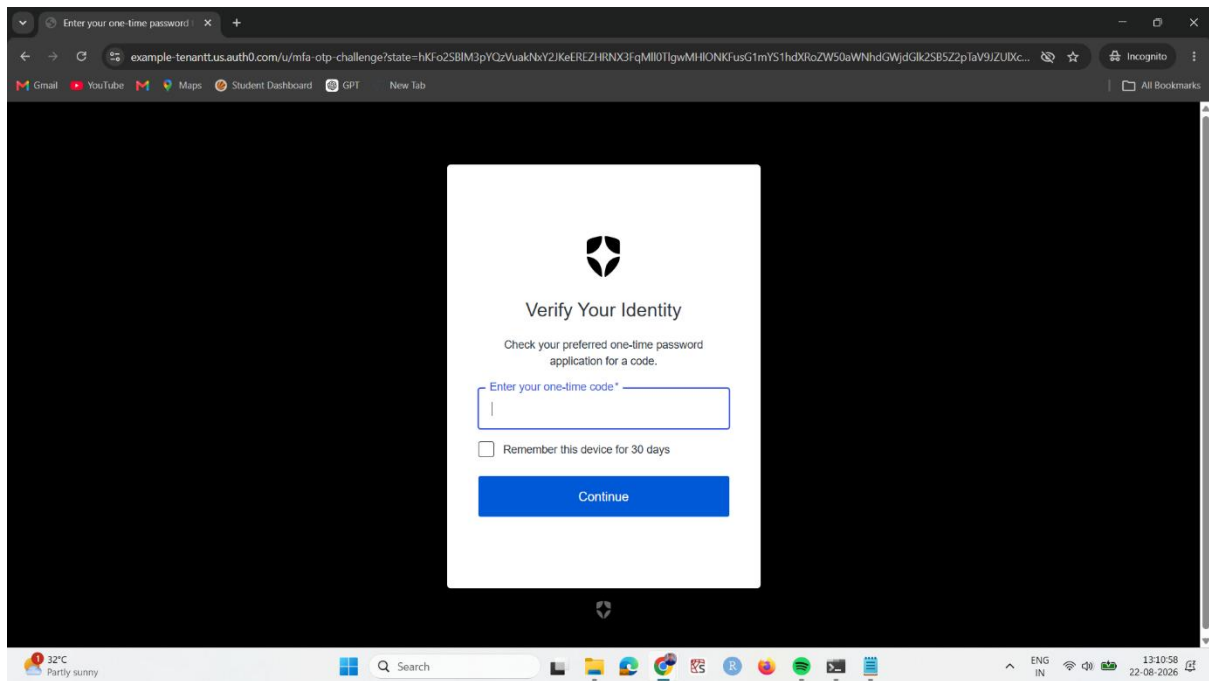
Screenshot 4 — Auth0 Universal Login: Custom Branding

Auth0 Universal Login page displaying the configured custom branding, demonstrating that the authentication experience has been customized for the AI application.



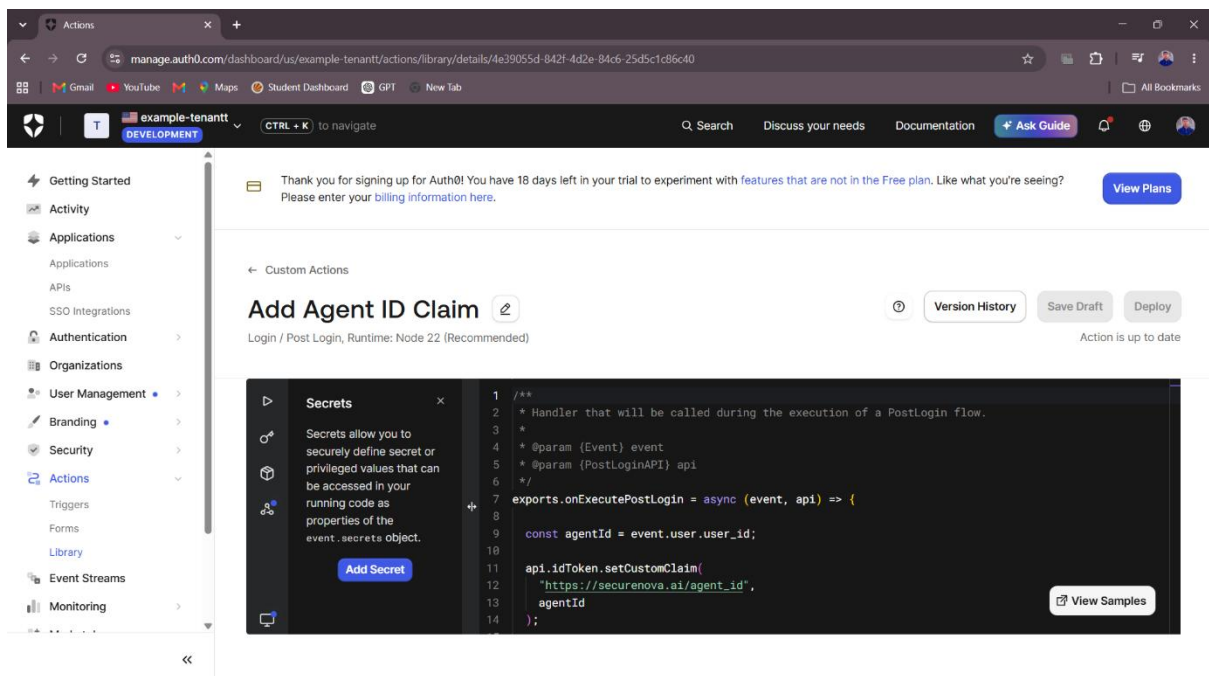
Screenshot 5 — Auth0 MFA: TOTP Enrollment/Challenge

Browser showing the Auth0 TOTP-based MFA enrollment or challenge flow, demonstrating that multi-factor authentication is enabled for users.



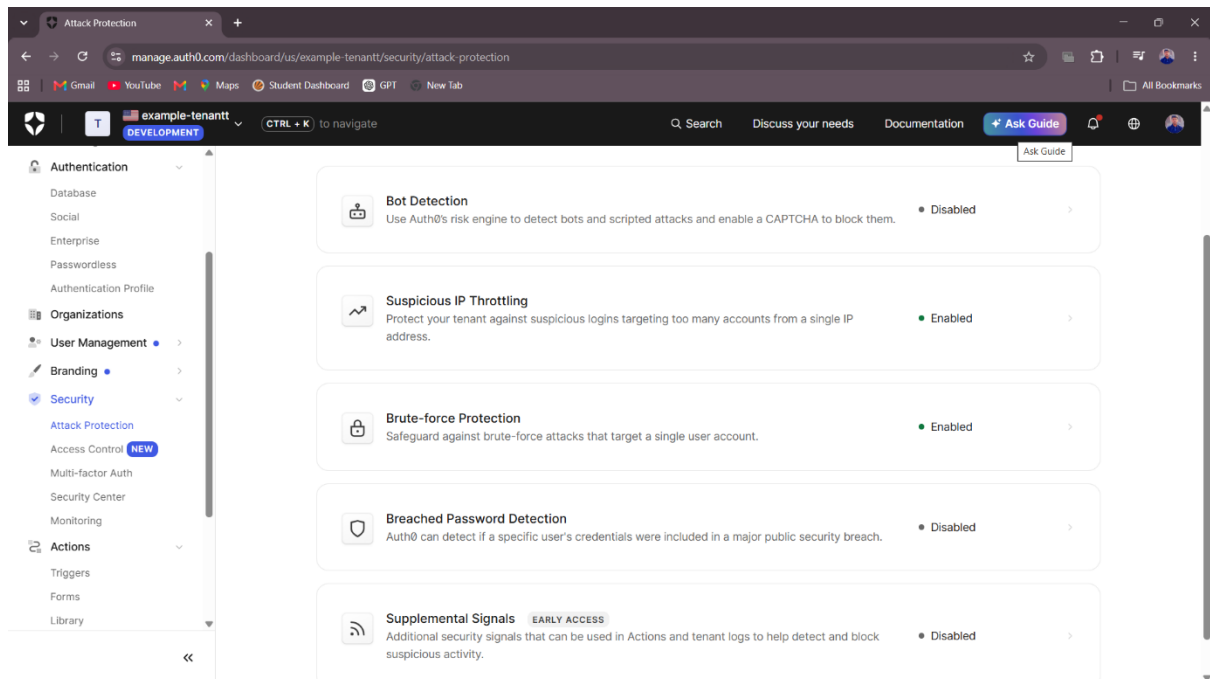
Screenshot 6 — Auth0 Actions: Post-Login Custom Claim

Auth0 Actions editor showing the post-login Action code that adds the `agent_id` custom claim to the user's ID token.



Screenshot 7 — Auth0 Attack Protection Configuration

Auth0 Attack Protection settings showing Brute-Force Protection and Suspicious IP Throttling enabled to protect the application against authentication attacks.



Screenshot 8 — Terminal: Credential Rotation and Token Replay

Timestamped terminal output showing the short-lived M2M credential successfully accessing the /chat endpoint with **200 OK**, followed by an expired-token replay returning **401 Unauthorized**.

```
Command Prompt
Using cached urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Collecting certifi>=2023.5.7 (from requests)
Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.5.1-cp314-cp314-win_amd64.whl (204 kB)
Using cached idna-3.19-py3-none-any.whl (68 kB)
Using cached urllib3-2.7.0-py3-none-any.whl (131 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Installing collected packages: urllib3, idna, charset_normalizer, certifi, requests
1/5 [idna] WARNING: The script idna.exe is installed in 'C:\Users\shiva\AppData\Roaming\Python\Python314\Script
s' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
2/5 [charset_normalizer] WARNING: The script normalizer.exe is installed in 'C:\Users\shiva\AppData\Roaming\Pyt
hon\Python314\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed certifi-2026.7.22 charset_normalizer-3.5.1 idna-3.19 requests-2.34.2 urllib3-2.7.0

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\shiva\securenova_ss8>python rotation.py
=====
SecureNova AI - Credential Rotation & Replay Test
=====
[2026-08-22 14:02:45] Using initial credential
[2026-08-22 14:02:47] First authenticated call -> 200 OK
Initial credential accepted

[2026-08-22 14:02:47] Rotating credential...
[2026-08-22 14:02:49] Credential rotated successfully

[2026-08-22 14:02:49] Replaying OLD credential...
[2026-08-22 14:02:51] Replay old credential -> 401 UNAUTHORIZED
OLD credential rejected successfully

=====
Credential rotation / replay protection test complete
=====

C:\Users\shiva\securenova_ss8>
```